

Laptop Price Prediction Using Regression Models

Analysis of Laptop Prices using Multiple Linear Regression and Proportional Odds Model

Project Report



Group 03

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1. Introduction

Laptops are widely used for education, business, software development, and personal computing. Differences in hardware specifications, design, and features lead to considerable variation in laptop prices, making it difficult for consumers to assess value. Statistical modeling provides an effective means of identifying the factors that influence laptop prices and developing predictive models to support purchasing, pricing, and marketing decisions.

1.1 About the Dataset

The dataset, obtained from the Kaggle platform, contains laptop prices and a range of hardware and software specifications from different manufacturers. It includes both categorical and continuous variables, making it suitable for regression analysis.

Source: Kaggle

<https://www.kaggle.com/datasets/muhammetvarl/laptop-price?resource=download>

1.2 Objectives of the Study

The main purpose of this study is to predict laptop prices using their specifications, including storage type, manufacturer, CPU, GPU, screen resolution, etc.

The specific objectives are:

- To perform descriptive statistical analysis and visualize the distribution of laptop prices and their relationship with explanatory variables.
- To examine the association between laptop specifications and market price.
- To build a Multiple Linear Regression model and a Proportional Odds Model for predicting laptop prices based on significant laptop characteristics.
- To evaluate the fitted regression model using appropriate diagnostic methods and verify that the model assumptions are reasonably satisfied.
- To interpret the effects of significant predictor variables on laptop prices and identify the most influential factors.

2. Literature Review

Guo and He (2024) used the Kaggle Laptop Price dataset, applying preprocessing, feature engineering, and regression models to predict prices from hardware specifications. Their results showed that processor type, RAM, storage, GPU, display specifications, and brand were the most influential predictors, and that careful feature engineering improved model performance by reducing redundancy. *Ballamudi (2025)* compared Linear Regression, Histogram Gradient Boosting, and XGBoost on 1,303 laptop records, evaluating performance using R^2 , MSE, RMSE, and MAE. XGBoost achieved the highest accuracy, while RAM and processor specifications were the strongest predictors, though the study confirmed that even simple linear regression captured meaningful price relationships. *Shaik et al. (2022)* and *Hansel et al. (2024)* both compared multiple linear regression against more flexible methods such as random forest, decision trees, and neural networks, consistently finding that linear regression performed competitively and offered clearer interpretability of how individual specifications relate to price. Hansel et al. further noted that laptop pricing reflects both interpretable linear relationships and non-linear patterns. *Turkolmez et al. (2024)* extended this by incorporating product age, condition, and component-level depreciation, highlighting pricing factors beyond standard specifications. Across these studies, hardware specifications—particularly processor type, RAM, storage, and GPU—consistently emerge as the most influential predictors of laptop price, and multiple linear regression repeatedly proves effective for modeling these relationships in an interpretable way. This supports the use of

multiple linear regression in the present study to build an explanatory model of continuous laptop prices using the Varlı dataset's specification variables.

3. Variable Transformation and Categorization

3.1 Variable definitions

Variable name	Variable description	Data Type
Company	Laptop Manufacturer (e.g., Dell, HP, Lenovo, Apple, Asus).	Categorical (19 levels)
Product	The brand or the model name of the laptop.	Categorical (618 levels)
TypeName	The type of the laptop such as Notebook, Ultrabook, Gaming, Workstation, or 2-in-1 Convertible.	Categorical (6 levels)
Inches	Screen size of the laptop measured diagonally in inches.	Continuous
ScreenResolution	Display resolution and screen technology (e.g., Full HD, IPS Panel, Touchscreen).	Categorical (40 levels)
Cpu	The processor model and specifications, including manufacturer, processor family, and clock speed.	Categorical (118 levels)
Ram	Installed Random Access Memory (RAM) capacity, measured in gigabytes (GB).	Continuous
Memory	Storage capacity and storage type, such as HDD, SSD, Hybrid, or Flash Storage.	Categorical (39 levels)
Gpu	Graphics Processing Unit (GPU) manufacturer and model used by the laptop.	Categorical (110 levels)
OpSys	Operating system installed on the laptop (e.g., Windows, macOS, Linux).	Categorical (9 levels)
Weight	Weight of the laptop is measured in kilograms (kg).	Continuous
Price	Market selling price of the laptop. This is the response (dependent) variable used in the regression analysis.	Continuous

3.2 Feature Engineering

Some categorical variables in the dataset had too many detailed categories. While these specific category levels provide very specific detailed information, including all of them in the model would have created many dummy variables. This would make the model complicated and difficult to explain. To address this, the detailed categories were grouped into broader, meaningful groups as follows,

Ram : Originally, the Ram variable contained the value with its units. Hence, the string ‘GB’ was removed and a numerical variable was created named ***Ram_GB*** to store the numerical values of the ***Ram*** Variable.

Weight : Originally, the Weight variable contained the value with its units. Hence, the string ‘kg’ was removed and a numerical variable was created named ***Weight_kg*** to store the numerical values of the ***Weight*** Variable.

OpSys : In the original dataset, there were 9 categories for this categorical variable. They were, “macOS”, “Mac OS X”, “Linux”, “Windows 10”, “Windows 10 S”, “Windows 7”, “Chrome OS”, “No OS”. Some of the categories were grouped and stored in a new variable ***OpSys_Group*** as follows,

1. “macOS” and “Mac OS X” were referring to the same operating systems “Mac”.
2. “Windows 10”, “Windows 10 S” and “Windows 7” were all grouped into one category named “Windows”.

Cpu : This category originally had 118 levels. The values were too detailed and might cause the model to overfit. Hence they were categorised into more reasonable categories and stored in a new variable ***Cpu_Group*** as follows,

New Category	Grouping Criterion
Intel Core i3	Matches any text containing "Intel Core i3"
Intel Core i5	Matches any text containing "Intel Core i5"
Intel Core i7	Matches any text containing "Intel Core i7"
Intel Core M	Matches any text containing "Intel Core M"
Intel Xeon	Matches any text containing "Intel Xeon"
Intel Celeron	Matches any text containing "Intel Celeron"
Intel Pentium	Matches any text containing "Intel Pentium"
Intel Atom	Matches any text containing "Intel Atom"
AMD A-Series	Matches the text "AMD A" followed immediately by one or more digits
AMD E-Series	Matches the text "AMD E" followed immediately by one or more digits
AMD Ryzen	Matches text containing "AMD Ryzen"
AMD FX	Matches text containing "AMD FX"
Other	Any processor string that fails to match every

	single one of the conditions listed above.
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Gpu : Originally the variable contained 110 levels. In a similar manner to the “Cpu Variable”, this variable also was too detailed and had too many levels. Therefore, it was categorized into more reasonable categories and stored in a new variable named **Gpu_Group**. The categories are as follows,

New Category	Grouping Criterion
Intel Iris	Matches text where "Intel" is followed somewhere later by "Iris"
Intel Integrated	Matches any text containing "Intel" that wasn't an Iris chip
AMD Radeon	Matches text where "AMD" is followed somewhere later by "Radeon", OR an "R" followed immediately by digits (like R5, R7), OR an "R" followed by digits and an "M" (like R7 M445).
Nvidia MX	Matches text where "Nvidia" is followed somewhere later by "MX"
Nvidia GTX	Matches text where "Nvidia" is followed somewhere later by "GTX"
Nvidia Quadro	"Nvidia" is followed somewhere later by "Quadro"
Nvidia Other	Matches any text containing "Nvidia" that didn't fit the GTX, MX, or Quadro criteria above
AMD FirePro	Matches text where "AMD" is followed somewhere later by "FirePro"
Other	Any row where the GPU string doesn't mention Nvidia, AMD, or Intel at all.

Memory : Originally the variable contained 39 levels. The variable contained information such as storage type and storage value.

i.e.,

“512GB SSD” - According to this information the laptop contains a SSD that has 512 gigabytes of storage.

Therefore, the variable information was divided into two meaningful variables, **Storage_Type** and **Total_Storage_GB**. **Storage_Type** contains information about the storage type(e.g., SSD, HDD, Flash,SSD+HDD, Hybrid, Flash+HDD, SSD+Hybrid). **Total_Storage_GB** contains the amount of storage the laptop has measured in gigabytes.

ScreenResolution : Originally the variable contained 40 levels. The variables contained information such as panel technology, screen resolution, whether it has a touchscreen or not. Hence the original variable was diffused into five new variables, **X_res** , **Y_res**, **Resolution_Group** , **Touchscreen**, **IPS**.

Resolution_Group variable contains information about the screen resolution. The values were categorised into four meaningful categories based on the following criteria. The groups were based on the width (given by *X_res*) and height (given by *Y_res*)

New Category	Criterion
QHD or Retina	$X_res \geq 2304$ and $Y_res \geq 1400$
4K	$X_res \geq 3840$ & $Y_res \geq 2160$
Full HD	$X_res \geq 1920$ & $Y_res \geq 1080$
HD	Everything else

Touchscreen variable is a binary variable pertaining to information whether there's a touchscreen or not. **IPS** variable is also a binary variable pertaining to information whether the screen uses IPS panel technology or not.

3.3 Interpretation of Newly Engineered Variables

Several new variables were created to improve data representation and model interpretation.

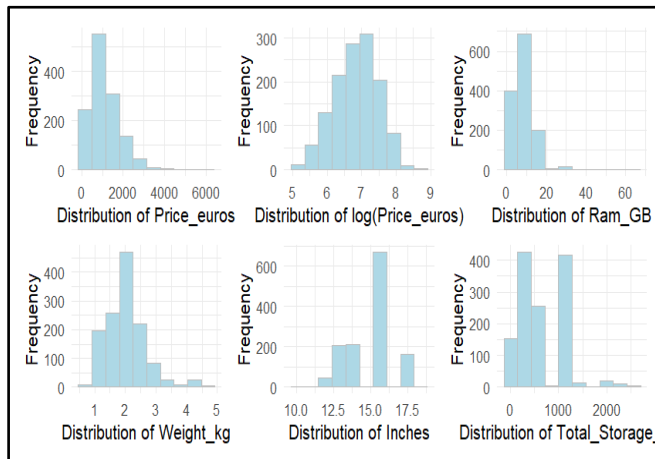
- **Ram_GB:** Installed RAM capacity (GB); higher RAM generally indicates better performance and higher price.
- **Weight_kg:** Laptop weight (kg); reflects portability and hardware configuration.
- **OpSys_Group:** Groups operating systems into Windows, Mac, Linux, Chrome OS, and Other for easier comparison.
- **Cpu_Group:** Groups processors into major families (e.g., Intel Core, AMD Ryzen) to represent processing performance.
- **Gpu_Group:** Groups graphics processors into major categories to capture differences in graphics capability.
- **Storage_Type:** Represents the storage technology (SSD, HDD, Hybrid, Flash, SSD + HDD), reflecting storage performance.
- **Total_Storage_GB:** Total storage capacity in gigabytes.
- **Touchscreen:** Indicates whether the laptop supports touchscreen functionality.
- **IPS:** Indicates whether the display uses IPS panel technology.
- **Resolution_Group:** Classifies display resolution into HD, Full HD, QHD/Retina, and 4K.
- **Company_Group:** Groups laptop manufacturers, combining rare brands into **Other**.
- **Price_Group:** Categorises laptop prices into **Low**, **Medium**, and **High** for the ordinal logistic regression model.

3.4 Variable Selection and Reduction for the Analysis

The original dataset contained several variables with numerous unique categories or overlapping information. To improve model interpretability and reduce dimensionality, several variables were grouped into broader categories, including **Company_Group**, **Cpu_Group**, **Gpu_Group**, **OpSys_Group**, **Storage_Type** and **Resolution_Group**. The original variables **Ram** and **Weight** were converted into the numerical variables **Ram_GB** and **Weight_kg**, while **Touchscreen**, **IPS** and **Total_Storage_GB** were derived from the original screen resolution and memory specifications. The original dataset included a variable **Product**, which contained information about the laptop model and model version. Due to its high cardinality (618 unique values), including this variable could cause the model to overfit. Therefore, the **Product** variable was excluded from the analysis. Although **OpSys_Group** was created during preprocessing, it was excluded from the regression models because the "Mac" operating system corresponded almost exclusively to "Apple" laptops, resulting in redundant information with **Company_Group**.

Following variable selection using forward and backward selection procedures, *Total_Storage_GB*, *Touchscreen*, and *IPS* were also excluded from the final regression models as they did not provide significant additional explanatory power. Therefore, the final analyses were performed using *Company_Group*, *TypeName*, *Inches*, *Ram_GB*, *Weight_kg*, *Cpu_Group*, *Gpu_Group*, *Storage_Type*, *Resolution_Group* and *Price_euros* and *Price_Group* for the ordinal logistic regression model.

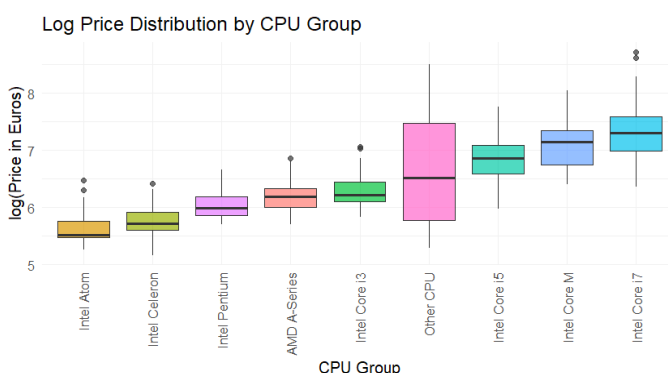
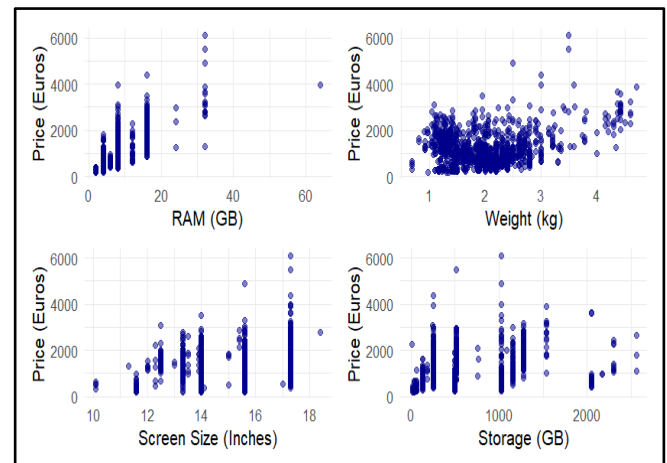
4. Descriptive Analysis



4.1 Exploratory Data Analysis

The histogram plots show the distribution of important laptop features. The laptop price, RAM, and storage variables show right-skewed distributions due to the presence of a few high-value laptops. The log transformation of price creates a more balanced distribution. Weight and screen size show more concentrated patterns with some variation. Overall, the histograms help identify the spread and skewness of the variables.

These scatter plots show how laptop price varies with RAM, weight, screen size, and storage capacity. A positive relationship is visible between RAM and price, indicating that laptops with more RAM are generally more expensive. Screen size also shows a slight positive trend with price. In contrast, weight and storage have weaker relationships, with prices spread across different values. Overall, RAM appears to have the strongest influence on laptop price among these features.

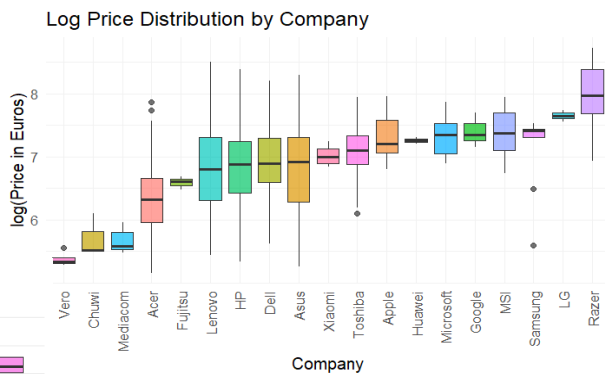
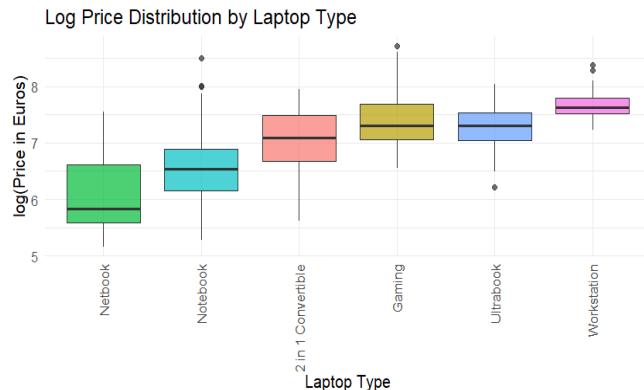


category shows a wide spread, indicating greater variation in prices. A few outliers are also present, representing laptops with unusually high prices within certain CPU groups.

This box plot compares the distribution of log-transformed laptop prices across different CPU groups. Laptops with Intel Core i7 and Intel Core i5 processors generally have higher median prices than other CPU types. Intel Atom and Intel Celeron processors are associated with lower-priced laptops. The "Other CPU"

This box plot shows the distribution of log-transformed laptop prices across different company groups. Premium brands such as Razer and LG tend to have the highest median prices, reflecting their focus on high-performance gaming and flagship devices, while budget brands such as Vero, Chuwi, and Mediacom are generally found at the lower end of the price range. Some companies, like Acer and Dell, display a much wider spread, indicating that these brands offer a broader range of laptops from budget

to premium models, whereas others, such as Huawei and MSI, show a narrower spread, suggesting more consistent pricing within their product lines. Outliers are also visible for brands like Acer and Samsung, suggesting the presence of a few unusually priced models relative to the rest of that brand's lineup.



This box plot compares the distribution of log-transformed laptop prices across different laptop types. Workstations have the highest median prices, followed by Gaming and Ultrabook laptops, indicating that these categories are generally more expensive due to their advanced hardware and performance. Netbooks have the lowest median prices, making them the most budget-friendly

option. The Notebook and 2-in-1 Convertible categories show moderate price ranges with some variation. Several outliers are also present, representing premium models that are priced significantly higher than the typical laptops in their category.

5. Multiple Linear Regression Model

5.1 Fitting the Multiple Linear Regression Model

5.1.1 Variable Selection and Model Building

To identify the most appropriate set of predictor variables, both backward elimination and forward selection methods were employed. In the backward elimination approach, the analysis began with the full model containing all candidate variables. Variables were then removed one at a time based on their statistical significance using the F-test.

For the forward selection approach, the analysis started with the null model and variables were added sequentially by comparing Akaike Information Criterion (AIC). Since several variables were statistically significant at different stages, **AIC was used as the main selection criterion to obtain a simpler model with better fit**. Variables were added sequentially, resulting in the inclusion of Cpu_Group, Ram_GB, TypeName, Storage_Type, Resolution_Group, Gpu_Group, Inches, Company_Group, and Weight_kg. The selection process stopped after adding Weight_kg, as adding Total_Storage_GB, Touchscreen, or IPS did not further improve the model according to AIC.

Both backward elimination and forward selection produced the same final regression model, indicating that Company_Group, TypeName, Inches, Ram_GB, Weight_kg, Cpu_Group, Gpu_Group, Storage_Type, and Resolution_Group were important predictors of laptop price, while Total_Storage_GB, Touchscreen, and IPS did not provide sufficient additional contribution to the final model.

5.2 Model Equation

$$\begin{aligned} \text{estimate(Price_euros)} = & \exp(6.7158 + 0.2250(\text{Apple}) + 0.0692(\text{Asus}) + 0.1249(\text{Dell}) + 0.1884(\text{HP}) + \\ & 0.0831(\text{Lenovo}) + 0.1730(\text{MSI}) + 0.2413(\text{Toshiba}) + 0.0950(\text{Other Company}) + \\ & 0.0949(\text{Gaming/Workstation}) - 0.1522(\text{Notebook/Netbook}) + 0.0951(\text{Ultrabook}) - 0.0470(\text{Inches}) + \\ & 0.0336(\text{Ram_GB}) + 0.0986(\text{Weight_kg}) - 0.3891(\text{Intel Low-End}) - 0.0156(\text{Intel Core i3}) + \end{aligned}$$

$0.3030(\text{Intel Core i5}) + 0.4041(\text{Intel Core i7}) + 0.2178(\text{Intel Core M}) + 0.0481(\text{Other CPU}) +$
 $0.1695(\text{Intel Integrated/Iris}) + 0.0516(\text{Nvidia GTX}) + 0.1284(\text{Nvidia MX}) + 0.4373(\text{Nvidia}$
 $\text{Quadro/Other}) + 0.4262(\text{Other GPU}) + 0.3336(\text{SSD} + \text{HDD}) + 0.0962(\text{HDD}) + 0.2322(\text{Hybrid}) +$
 $0.2748(\text{SSD}) - 0.2822(\text{Full HD}) - 0.3902(\text{HD}) - 0.0739(\text{QHD or Retina})$

5.3 Interpretation of the model parameters

The intercept of **6.7158** represents the expected log price of a laptop belonging to all reference categories with zero values for the continuous predictors. Since such a laptop is not realistic, the intercept mainly serves as the baseline of the regression equation.

Company: Using **Acer** as the reference category, Apple, Toshiba, HP, MSI, Dell, Lenovo, Asus and Other brands are associated with approximately **25.2%, 27.3%, 20.7%, 18.9%, 13.3%, 8.7%, 7.2% and 10.0%** higher prices, respectively, indicating that manufacturer has a substantial influence on laptop price.

Laptop Type: Compared with **2-in-1 Convertible** laptops, **Gaming, Workstation and Ultrabook** models are generally associated with higher prices, whereas **Notebook** and **Netbook** models are associated with lower prices after adjusting for the remaining predictors.

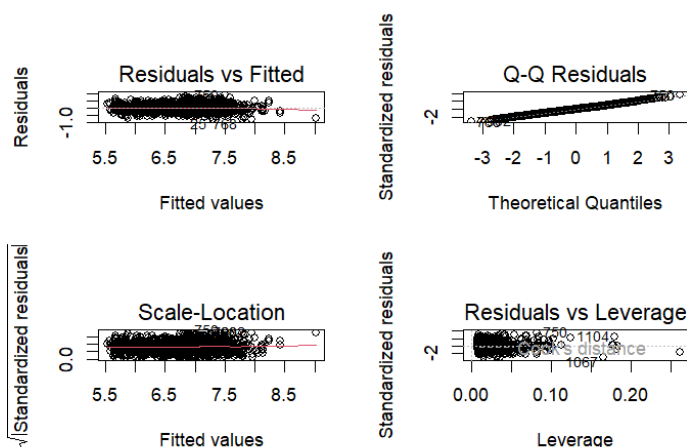
Continuous Variables: Holding all other variables constant, each additional **1 GB of RAM** increases the expected laptop price by approximately **3.4%**, while each additional **1 kg of weight** increases price by approximately **10.4%**. Conversely, a one-inch increase in screen size is associated with an estimated **4.6% decrease** in expected price.

CPU and GPU: Relative to the AMD reference group, **Intel Core i5, Intel Core i7 and Intel Core M** processors are associated with higher laptop prices, while **Intel Low-End** processors are associated with lower prices. Similarly, **Nvidia Quadro/Other professional GPUs** show the largest positive price premium, followed by Nvidia MX, Nvidia GTX and Intel Integrated/Iris graphics.

Storage Type: Using **Flash Storage** as the reference category, **SSD + HDD, SSD, Hybrid** and **HDD** storage configurations are associated with approximately **39.6%, 31.6%, 26.1% and 10.1%** higher laptop prices, respectively.

Screen Resolution: Using **4K Ultra HD** as the reference category, **Full HD, HD** and **QHD/Retina** displays are associated with approximately **24.6%, 32.3% and 7.1%** lower prices, respectively, indicating that higher display resolution is generally associated with higher laptop prices.

5.4 Residual Analysis



• Residuals vs Fitted Plot

The residuals are randomly scattered around the zero line without any strong systematic pattern, indicating that the linear regression model provides an adequate fit to the data. The red smoothing line shows only a slight downward trend at higher fitted values, suggesting a minor departure from perfect linearity. A few observations appear to be outliers; however, they are limited in number and do not appear to substantially affect the overall model. Therefore, the linearity assumption is reasonably satisfied.

• Q-Q Residuals Plot

The Q-Q plot indicates that most residuals closely follow the reference line, suggesting that the residuals are approximately normally distributed. Slight deviations from the line are evident at both tails, indicating the presence of a few extreme observations. Since the majority of points align well with the reference line, the normality assumption is considered reasonably satisfied.

• Scale-Location Plot

The Scale-Location plot shows that the residuals are fairly evenly dispersed across the fitted values, with no strong systematic pattern. The red smoothing line exhibits a slight upward trend, indicating a small increase in residual variance as the fitted values increase. This suggests mild heteroscedasticity; however, the change in variance is not substantial. Overall, the assumption of constant variance is reasonably satisfied.

• Residuals vs Leverage Plot

Most observations have low leverage and are clustered near the origin, indicating that the majority of the data points have little influence on the fitted regression model. A small number of observations exhibit relatively higher leverage; however, they remain within the Cook's Distance boundaries. Consequently, there is no evidence of highly influential observations that would substantially affect the regression results.

Overall Assessment

Overall, the diagnostic plots indicate that the assumptions of **linearity, normality, homoscedasticity, and the absence of influential observations** are reasonably satisfied. While a few minor deviations and potential outliers are present, they are not substantial enough to affect the validity of the model. Therefore, the fitted multiple linear regression model can be considered appropriate for predicting laptop prices.

6. Ordinal Logistic Regression Model

6.1 Fitting the Logistic Regression Model

6.1.1 Variable selection and Model Building

For this model, the response variable *Price_euros* was categorized into three meaningful groups using an automated statistical approach called “**tertile split segmentation**”, which uses the dataset's internal distribution to slice the laptops into three equally sized economic tiers.

To achieve this balanced grouping, the complete range of laptop prices was scanned to identify the specific 33.3rd and 66.6th percentile boundaries that define the thresholds of the data. Each laptop was then systematically sorted into its respective mathematical bracket based on these cut-off points, ensuring that the lowest-cost hardware was safely captured within the baseline group. Finally, by mapping these buckets to the labels “*Low*”, “*Medium*” and “*High*” the data was formally transformed into a true ordinal factor where the natural rank structure (Low < Medium < High) is explicitly preserved.

This distribution-driven approach guarantees that each economic tier contains a virtually identical number of observations, mitigating class imbalance and providing a highly stable, uniform baseline configuration for the downstream ordinal logistic regression framework.

To find the most suitable predictor variables for the model, the Forward Selection Method was used. The selection method produced the same set of predictors that was used for the MLR model. Which included *Company_Group*, *TypeName*, *Inches*, *Ram_GB*, *Weight_kg*, *Cpu_Group*, *Gpu_Group*, *Storage_Type*, *Resolution_Group*.

6.2 Model Equation

The fitted cumulative logit model is

$$\log \left(\frac{P(Y \leq j)}{P(Y > j)} \right) = \alpha_j - \eta, \quad j = 1, 2,$$

where,

$$\begin{aligned} \eta = & 0.29615(\text{Ram_GB}) + 2.27469(\text{Apple}) + 1.04452(\text{Asus}) + 1.63319(\text{Dell}) + 2.02117(\text{HP}) + \\ & 1.25431(\text{Lenovo}) + 1.50229(\text{MSI}) + 2.63661(\text{Toshiba}) + 1.74034(\text{Other Company}) - 1.25478(\text{Intel} \\ & \text{Low-End}) + 0.28476(\text{Intel Core i3}) + 3.03213(\text{Intel Core i5}) + 3.90724(\text{Intel Core i7}) + 2.41938(\text{Intel} \\ & \text{Core M}) + 2.75208(\text{Other CPU}) + 1.40932(\text{Intel Integrated/Iris}) + 0.75654(\text{Nvidia GTX}) + \\ & 0.43277(\text{Nvidia MX}) + 2.34497(\text{Nvidia Quadro/Other}) + 2.30862(\text{Other GPU}) + 2.27123(\text{SSD} + \\ & \text{HDD}) + 0.51671(\text{HDD}) + 1.53592(\text{Hybrid}) + 1.80895(\text{SSD}) - 2.08118(\text{Full HD}) - 3.03462(\text{HD}) - \\ & 1.37137(\text{QHD or Retina}) - 0.30289(\text{Inches}) \end{aligned}$$

The threshold (cut-point) parameters are ,

$$\begin{aligned} \alpha_1 &= 1.0572 & (\text{Low}|\text{Medium}), \\ \alpha_2 &= 3.9498 & (\text{Medium}|\text{High}). \end{aligned}$$

Because **TypeName** was specified as a **nominal effect**, separate adjustments are estimated for each threshold.

For the **Low | Medium** threshold,

$$\alpha_1^* = 1.0572 - 2.6450(\text{Gaming/Workstation}) - 0.1405(\text{Notebook/Netbook}) - 1.8435(\text{Ultrabook})$$

For the **Medium | High** threshold,

$$\alpha_2^* = 3.9498 - 0.2951(\text{Gaming/Workstation}) + 1.1384(\text{Notebook/Netbook}) - 0.3293(\text{Ultrabook}).$$

Final fitted equations

For the probability of belonging to the **Low** price category or below,

$$\log \left(\frac{P(Y \leq \text{Low})}{P(Y > \text{Low})} \right) = \alpha_1^* - \eta$$

and for the probability of belonging to the **Medium** price category or below,

$$\log \left(\frac{P(Y \leq \text{Medium})}{P(Y > \text{Medium})} \right) = \alpha_2^* - \eta$$

where **Acer**, **AMD CPU**, **AMD GPU**, **Flash Storage**, and **4K Ultra HD** are the reference categories.

6.3 Interpreting model parameters

RAM

The coefficient for **RAM (GB)** is **0.2962**, indicating that each additional gigabyte of RAM increases the log-odds of a laptop belonging to a higher price category by 0.2962. The corresponding odds ratio is 1.34, implying that each additional gigabyte of RAM increases the odds of being in a higher price category by approximately **34.5%**, holding all other variables constant.

Company Group

Using **Acer** as the reference category, Apple laptops have a coefficient of **2.2747**, corresponding to an odds ratio of approximately **9.73**. This indicates that Apple laptops have about **9.7 times greater odds** of being classified into a higher price category than Acer laptops, after controlling for the other predictors. Similarly, Asus, Dell, HP, Lenovo, MSI, Toshiba, and other laptop brands have odds ratios of approximately **2.84**, **5.12**, **7.55**, **3.51**, **4.49**, **13.97**, and **5.70**, respectively. These results suggest that Toshiba, Apple, and HP laptops are considerably more likely to belong to higher price categories than Acer laptops.

CPU Group

Using **AMD processors** as the reference category, laptops equipped with Intel Low-End processors and Intel Core i3 processors do not differ significantly from AMD-based laptops in their odds of belonging to a higher price category. In contrast, Intel Core i5, Intel Core i7, Intel Core M, and other processor types significantly increase the likelihood of belonging to a higher price category. Their estimated odds ratios are approximately **20.74**, **49.76**, **11.24**, and **15.67**, respectively. Among these processor categories, Intel Core i7 exhibits the strongest positive association with higher-priced laptops.

GPU Group

Using **AMD graphics processors** as the reference category, laptops equipped with Intel Integrated or Iris graphics have approximately **4.09 times greater odds** of belonging to a higher price category. Nvidia GTX graphics increase the odds by approximately **2.13 times**, although this effect is only marginally significant ($p = 0.079$). Nvidia MX graphics do not differ significantly from AMD graphics. Laptops equipped with Nvidia Quadro or other professional graphics processors have approximately **10.43 times greater odds** of belonging to a higher price category, while other GPU types have approximately **10.06 times greater odds** than AMD graphics processors.

Storage Type

Using **Flash Storage** as the reference category, laptops equipped with both SSD and HDD storage have approximately **9.69 times greater odds** of belonging to a higher price category. Similarly, laptops with SSD storage alone have approximately **6.10 times greater odds** of being classified into a higher price category than those with Flash Storage. Although laptops with HDD and Hybrid storage have estimated odds ratios of approximately **1.68** and **4.65**, respectively, these effects are not statistically significant, indicating insufficient evidence that these storage types differ from Flash Storage in determining price category.

Resolution Group

Using **4K Ultra HD** displays as the reference category, laptops with Full HD displays have an odds ratio of approximately **0.125**, indicating that they are about **87.5% less likely** to belong to a higher price category than laptops with 4K Ultra HD displays. Likewise, HD displays have an odds ratio of approximately **0.048**, meaning that they are about **95.2% less likely** to belong to a higher price category than 4K Ultra HD displays. Although laptops with QHD or Retina displays have lower estimated odds (odds ratio approximately **0.254**), this effect is not statistically significant.

Inches

The coefficient for **screen size (Inches)** is **-0.3029**, indicating that each additional inch of screen size decreases the log-odds of belonging to a higher price category. The corresponding odds ratio is approximately **0.739**, implying that each one-inch increase in screen size decreases the odds of belonging to a higher price category by approximately **26.1%**, after accounting for the remaining variables.

Type of Laptop (Nominal Effects)

The variable **TypeName** violates the proportional odds assumption and has therefore been modelled as a **nominal effect**, meaning that its influence differs across the price thresholds. Consequently, a single odds ratio cannot be reported for each laptop type. Instead, the estimated threshold coefficients indicate

that the effect of laptop type varies between the transition from **Low to Medium** and the transition from **Medium to High** price categories.

For the **Low versus Medium** threshold, Gaming/Workstation laptops have a significant negative coefficient (**-2.645**, $p < 0.01$), indicating that their effect on moving out of the Low price category differs substantially from the reference laptop type. Ultrabooks also show a significant negative effect (**-1.844**, $p < 0.05$), whereas Notebook/Netbook models do not differ significantly from the reference category.

For the **Medium versus High** threshold, only Notebook/Netbook laptops exhibit a statistically significant effect (**1.138**, $p < 0.001$), indicating that their influence differs at the upper price threshold compared with the reference laptop type. Gaming/Workstation and Ultrabook categories are not statistically significant at this threshold.

Threshold Coefficients

The threshold coefficients (**Low|Medium** and **Medium|High**) define the cut-off points separating the three ordered price categories. These parameters are necessary for estimating the cumulative probabilities of belonging to each price category but do not have a direct practical interpretation in terms of the effects of the predictor variables. Consequently, they are typically reported but not discussed in detail in the interpretation of the model.

6.4 Model adequacy checking

The adequacy of the fitted ordinal logistic regression model was evaluated using likelihood ratio tests, Akaike Information Criterion (AIC), and classification performance measures. The likelihood ratio test comparing the final model with the intercept-only (null) model yielded a test statistic of **1621.5** with **34 degrees of freedom** and a **p-value** $< 2.2 \times 10^{-16}$, indicating that the inclusion of the predictor variables significantly improves the model fit over a model with no explanatory variables. Furthermore, the comparison between the standard proportional odds model and the partial proportional odds model, in which **TypeName** was modelled as a nominal effect, resulted in a likelihood ratio statistic of **10.664** with **3 degrees of freedom** (**p = 0.0137**). This significant result suggests that relaxing the proportional odds assumption for **TypeName** provides a statistically significant improvement in model fit. The final model achieved an **AIC of 1313.435**, which is lower than that of the standard proportional odds model (**AIC = 1318.1**), indicating that the selected model provides a better balance between goodness-of-fit and model complexity.

The predictive performance of the model further demonstrates its adequacy. The confusion matrix shows that the model correctly classified **370 of 437 Low-priced laptops (84.7%)**, **301 of 432 Medium-priced laptops (69.7%)**, and **345 of 434 High-priced laptops (79.5%)**. Overall, the model correctly classified **77.97%** of the **1,303** observations, indicating good predictive performance across the three price categories. In addition, the **Mean Absolute Error (MAE) of 0.2218** indicates that the average prediction error is less than one-quarter of a price category, suggesting that misclassifications are generally close to the true category. Examination of the confusion matrix also reveals that nearly all classification errors occur between adjacent categories (Low–Medium or Medium–High), with virtually no direct misclassification between the Low and High categories. This pattern is desirable in ordinal classification because it demonstrates that the model preserves the natural ordering of the response variable. Overall, these results indicate that the partial proportional odds model provides an adequate fit to the data and possesses good discriminative ability for predicting laptop price categories.

7. Discussion

Findings

The multiple linear regression (MLR) model identified **Company Group, TypeName, Inches, Ram_GB, Weight_kg, CPU Group, GPU Group, Storage Type, and Resolution Group** as significant predictors of laptop price, explaining **82.8%** of the variation in the logarithm of laptop prices (Adjusted $R^2 = 82.8\%$). Higher RAM capacity, weight, premium CPU and GPU categories, SSD-based storage configurations, and premium laptop brands were associated with higher prices, whereas larger screen sizes and lower display resolutions were associated with lower prices.

The ordinal logistic regression (OLR) model similarly identified **Ram_GB, Company Group, CPU Group, GPU Group, Storage Type, Resolution Group, and Inches** as significant predictors of laptop price category. The partial proportional odds model provided a significantly better fit than the standard proportional odds model, indicating that **TypeName** violated the proportional odds assumption. The model achieved a **classification accuracy of 77.97%**, demonstrating good predictive performance for classifying laptops into Low, Medium, and High price categories.

Summary

Both modelling approaches consistently identified hardware specifications and manufacturer brand as the primary determinants of laptop pricing. The MLR model effectively explained variations in continuous laptop prices, while the OLR model accurately classified laptops into ordered price categories. Together, these models provide complementary insights into the factors influencing laptop prices.

Conclusion

The study demonstrates that laptop prices are primarily influenced by RAM capacity, processor type, graphics card, storage configuration, display resolution, manufacturer brand, and screen size. The MLR model exhibited strong explanatory power, while the OLR model showed good classification performance. Overall, the two modelling approaches provide a comprehensive understanding of the factors affecting laptop pricing and offer useful tools for both price prediction and price category classification.

Challenges and Limitations

Although both models performed well, the MLR model showed slight departures from the assumptions of normality and homoscedasticity, while the OLR model required a partial proportional odds specification because the proportional odds assumption was violated for **TypeName**. In addition, some categorical levels contained relatively few observations, which may have reduced the precision of certain coefficient estimates. The dataset also lacked variables such as battery life, processor generation, release year, build quality, warranty, and market demand, which could further improve model performance in future studies.

8. References

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9. Member Contribution

Name	Index	Contribution
G M S H De Silva	S17226	Data Pre-processing Model Fitting(for both models) Report Writing Model Adequacy Checking Exploratory Data Analysis Model Interpretation
K.H.S.B. Kumarasinghe	S17250	Data Pre-processing Feature Engineering Model Fitting(for both models) Report Writing Literature Review
Keshine Kamalakkannan	S17409	Model Fitting(for both models) Literature Review Report Writing Model Adequacy Checking Model Interpretation
Pratheeka Thushyanthan	S17435	Variable Selection for MLR using Forward Elimination Literature Review Proof-Reading
Sajani Wathsala	S17441	Variable Selection for MLR using Backward Elimination Report Writing Literature Review Exploratory Data Analysis